

Vehicle Stability Control for 3-Wheeler EV

(Independent Rear Hub Motors)

Team Analog

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The Engineering Challenge

Transitioning to Independent Rear Hub Motors removes the mechanical axle, creating three critical safety gaps:

1. Loss of Differential Action

Without a mechanical link, there is no physical mechanism to ensure the outer wheel spins faster than the inner wheel during a turn.

Risk: *Severe tire scrubbing and inability to steer.*

2. Inherent Instability (Rollover Risk)

Commercial 3-wheelers have a triangular footprint and high Center of Gravity (CG). Without active control, the vehicle is prone to tipping over during sharp cornering.

Risk: *Catastrophic rollover events.*

3. Variable Payload Dynamics

Empty Weight: ≈ 400 kg vs. **Loaded Weight:** ≈ 900 kg.

Risk: *A control system tuned for one scenario often causes instability in the other.*

Design Objective

To engineer a robust **"Active Electronic Differential"** that guarantees stability, traction, and safety across all payloads—without adding expensive sensors.

Features: Active Rollover Prevention (PLTR) & Regenerative Torque Vectoring

To replace the mechanical differential while ensuring safety under variable commercial payloads, we propose a **Hierarchical Control Architecture**. A central Vehicle Control Unit (VCU) handles dynamics planning, while distributed Motor Controllers execute high-frequency traction tasks.

- **Steering Angle Sensor (SAS):** Mounted on the column. Outputs steering angle (δ_f) for trajectory planning, and the VCU calculates the rate of change ($\dot{\delta}_f$) for predictive rollover detection.
- **Pedal Position Sensors:** Maps driver intent to T_{total} (Total Torque Request). Aggregates positive throttle ($+T_{req}$) and negative regenerative braking ($-T_{req}$) demands.

- **Inertial Measurement Unit (IMU):** A Bosch BMI088 (6-axis) mounted at the chassis Center of Gravity (CG). Outputs actual yaw rate (ω_{actual}) and lateral acceleration (a_y) for stability and rollover detection.
- **Wheel Speed Sensors:** High-resolution encoders on each driven rear hub output ω_{wheel} for real-time Slip Ratio (λ) calculations. **Crucially, a wheel speed sensor is also located on the un-driven front wheel** to accurately derive the true forward vehicle velocity (v_x). This prevents false velocity readings if both rear wheels lose traction simultaneously.

- **Function:** Calculates the Target Yaw Rate (ω_{des}) based on steering input and vehicle speed, accounting for tire slip angles that develop at higher velocities.

$(K_{us} = \text{understeer gradient})$

- *Danger Mode (PLTR > 0.8)*: If impending lift is predicted, the controller overrides steering logic and commands a **massive negative braking torque on the outer wheel** to stabilize the chassis.



Layer 3: Torque Allocator (The "Virtual Differential")

Converts the calculated Yaw Moment (M_z) into specific torque commands for the left and right motors.

- **Logic:** Superimposes steering effort onto the driver's requested torque (T_{total}).
- **Formulas:** $T_{Right} = \frac{T_{total}}{2} + \Delta T$ and $T_{Left} = \frac{T_{total}}{2} - \Delta T$ where $\Delta T = \frac{M_z \cdot r_{tire}}{W_{track}}$
- **Note:** During braking ($T_{total} < 0$), this automatically performs **Regenerative Vectoring**.

Block D: Local Slip Control Loop (Traction Execution)

Embedded directly in the Motor Controllers (FOC), this high-frequency loop ensures traction.

- **Action:** Acts as a "Ceiling Limiter." By comparing driven wheel speeds against true velocity (v_x) from the front un-driven wheel, if $\lambda > 15\%$ (loss of traction), it overrides the VCU torque command and cuts current locally to re-adhere the tire.

4 Qualitative Performance Analysis

1. Longitudinal Performance: Active Traction Optimization

Unlike a mechanical open differential, which sends equal torque to both wheels regardless of grip, our **Block D Slip Limiter** ensures continuous acceleration.

- **Launch Acceleration:** By continuously monitoring Slip Ratio (λ), the system maintains wheel slip within the optimum window of **10–15%**. This is the physical peak of the friction curve where maximum traction occurs.
- **Surface Independence:** On split-friction surfaces (e.g., one wheel on tarmac, one on dirt), the controller independently throttles the slipping wheel while maintaining full torque on the gripping wheel.

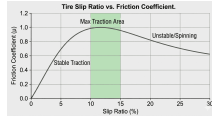


Figure 2: Target 10-15% Slip Region

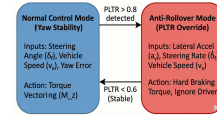


Figure 3: Rollover Safety Logic

2. Lateral Stability Steering Behavior

The **Layer 2 SMC** actively manages the vehicle's yaw dynamics to solve 3-wheeler instability.

- **Turn-In Response:** Upon entering a turn, the Torque Allocator generates a positive yaw moment ($+M_z$) by over-driving the outer wheel, eliminating "understeer/plowing."
- **Rollover Prevention:** The system continuously calculates the **Predictive Lateral Load Transfer Ratio (PLTR)**.
 - **Threshold:** If $PLTR > 0.8$ (imminent lift predicted), the SMC transitions to a safety state.
 - **Action:** Commands massive negative braking torque on the outer wheel to realign the chassis and widen the turning radius before an un-tripped rollover can initiate.

3. Combined-Slip Braking Stability (Electronic EBD)

During "Braking-While-Turning" maneuvers, the system acts as an active **Electronic Brake Force Distributor**.

- **Vectoring:** The Allocator shifts regenerative load to the inner wheel to create a pivoting moment, helping the vehicle turn in while slowing down.
- **Anti-Lock Logic:** Simultaneously, the Slip Control Loop prevents wheel lock-up. If the vectoring demand exceeds the tire's friction circle, the controller reduces regenerative torque to restore lateral steering authority.

5 Design Feasibility Constraints

Our proposed architecture directly addresses the operational constraints of a commercial mass-market vehicle:

Constraint	Challenge	Our Control Solution
Mass Variation	Payload changes (I_z) cause PID loops to fail.	SMC (Layer 2) utilizes a "Switching Gain" tuned for max inertia, providing robustness against 40% mass changes without adaptive tuning.
Sensor Cost	Sideslip/Torque sensors are too expensive.	Kinematic Observation: We estimate states using sensor fusion from the low-cost Bosch BMI088 IMU , and standard Hall-effect wheel encoders utilizing the front un-driven wheel as a true velocity reference.
Efficiency	ESP typically wastes energy by friction braking.	Regenerative Vectoring: The Allocator prioritizes regenerative braking on the inner/outer wheels during cornering or correction, recovering kinetic energy while stabilizing the turn.